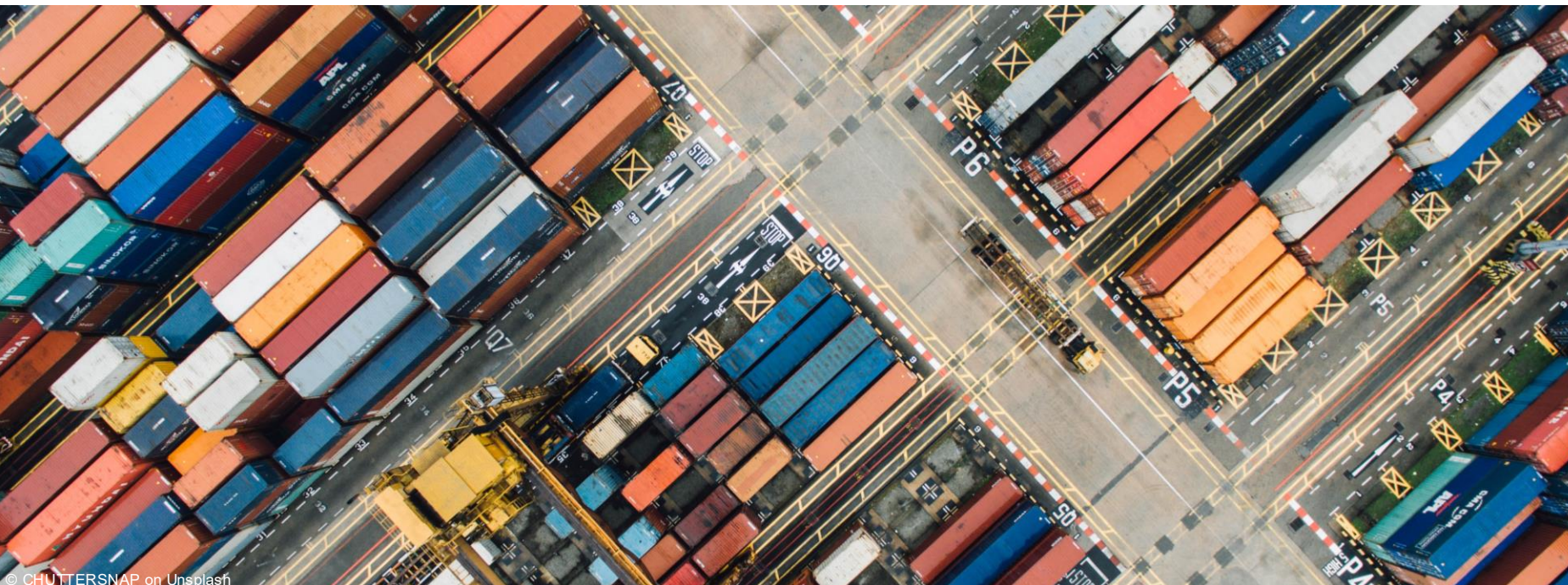


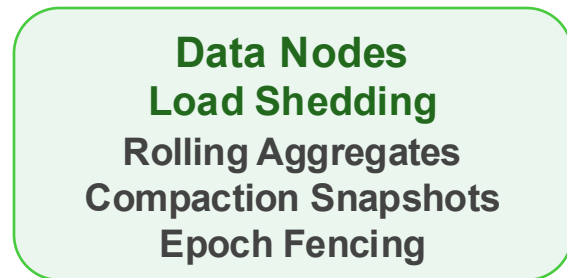


StreamShard: A Distributed Event Aggregation Engine

Jakob Al Khuzayi & Javier Herrero Delgado | Scalability Engineering | July 13, 2026

<https://github.com/Jakob-al28/StreamShard>





Demo: Primary Failover and Live Resharding

```
[jakob@archlinux streamshard]$ ./demo/demo.sh up
building...

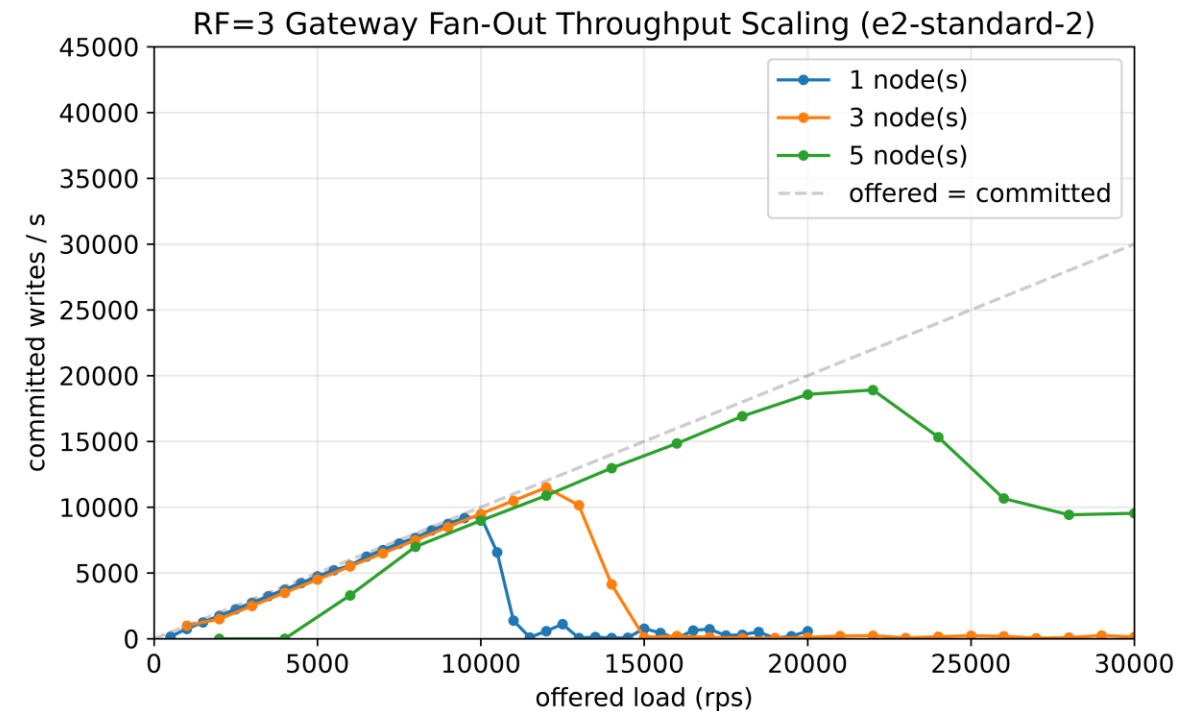
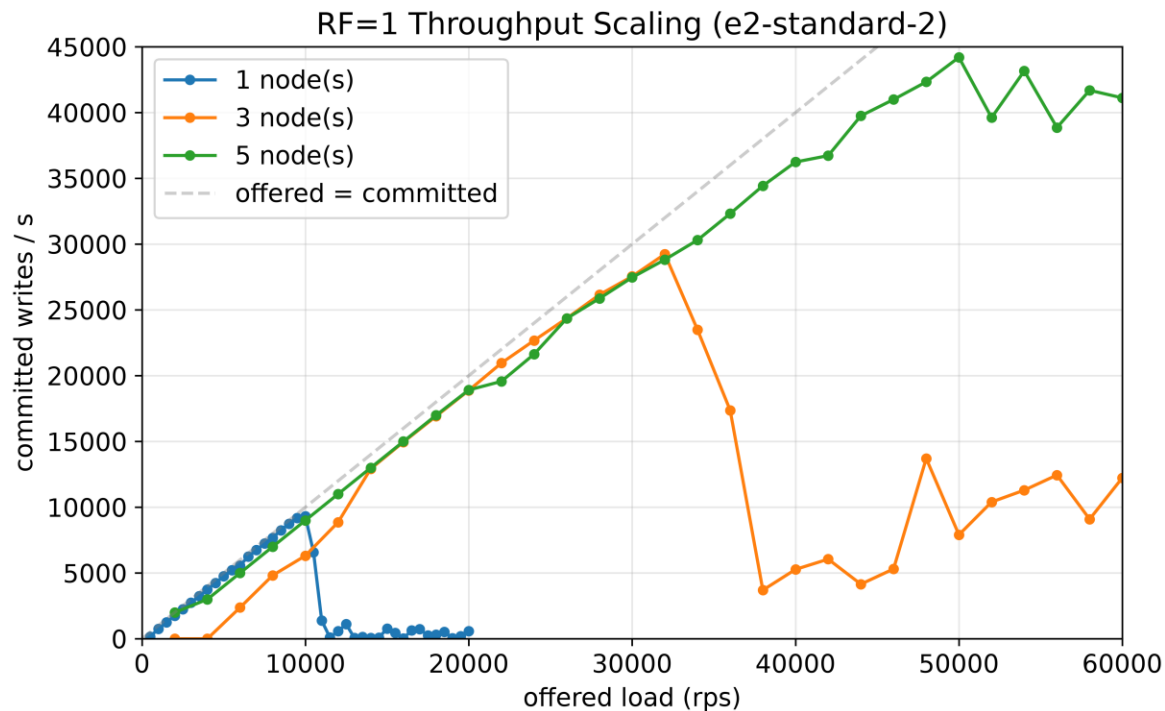
cluster up: CP=127.0.0.1:6060 GW=127.0.0.1:7070 nodes=127
.0.0.1:8081 127.0.0.1:8082 127.0.0.1:8083 RF=3 W=2
primary for 'demo-key' = 127.0.0.1:8083 (node2, pid 105870
2)

[jakob@archlinux streamshard]$ ./demo/demo.sh load
```

```
[jakob@archlinux streamshard]$
```

Scalability Results: The Cost of Replication

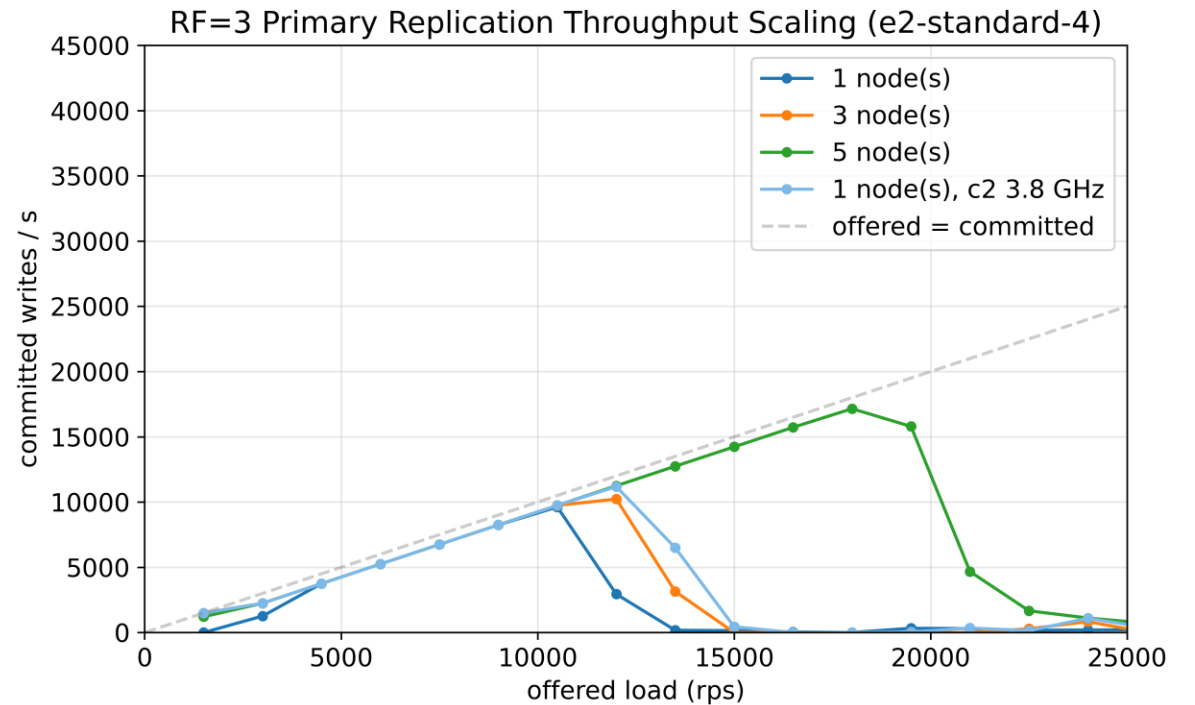
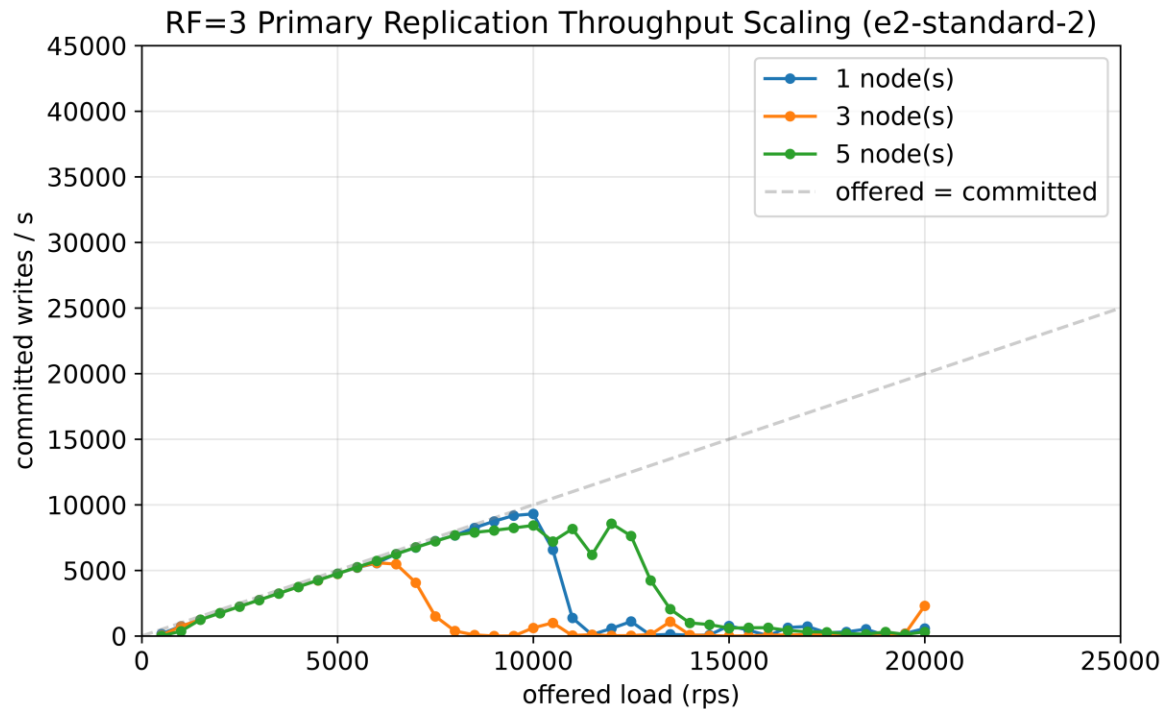
Write Quorum replication (RF=3, W=2) reduces peak throughput by **~60%**



Scalability Results: Leader-Based Replication & Vertical Scaling

Primary-driven replication costs **~55%** throughput vs. gateway fan-out

Core count helps parallel work, clock speed helps serial work

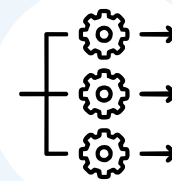


System Limitations & Future Work



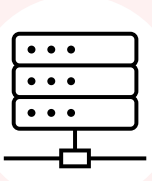
Per-node global lock on the WAL

Serialized commits limit throughput at **~10,000 writes/s** per node



Scale Up & Parallel Write:

Shard partition by key across workers



Centralized Control Plane

Single coordinator manages epochs and resharding



Replicated Control Plane

backed by a consensus store (e.g. ZooKeeper, etcd).

See the *README* for a full list of known limitations.